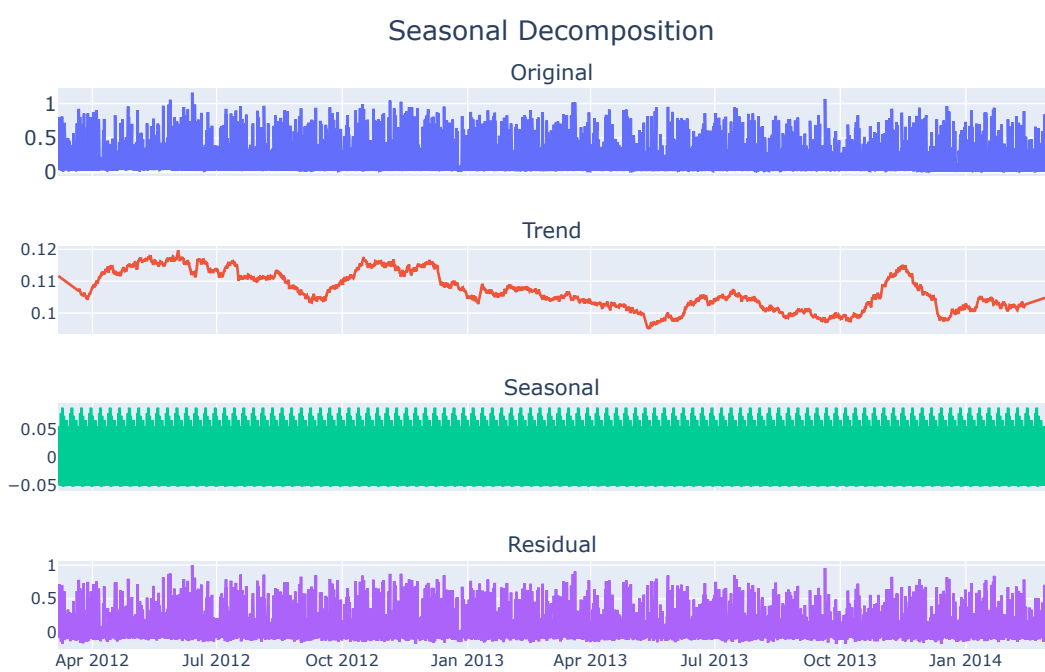


Seasonal Decomposition

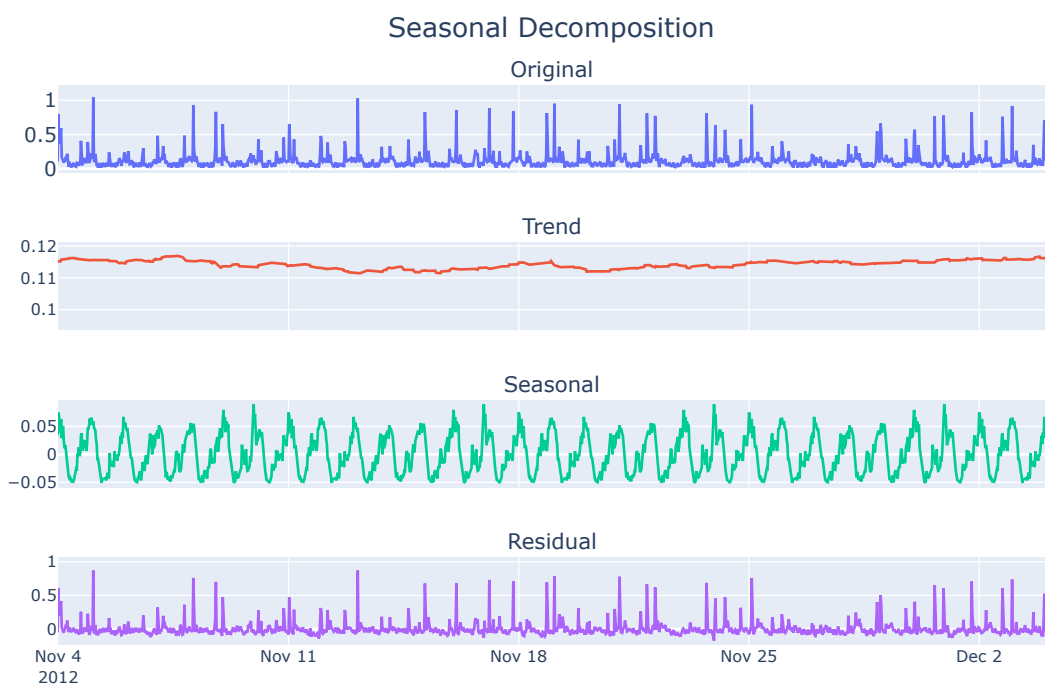
```
ts = SeasonalInterpolation(seasonal_period=48*7).fit_transform(ts_df.
energy_consumption.values.reshape(-1,1)).squeeze()
```

Seasonal Decomposition

```
#Does not support misssing values, so using imputed ts instead
res = seasonal_decompose(ts, period=7*48, model="additive",
extrapolate_trend="freq", filt=np.repeat(1/(30*48), 30*48))
fig = decomposition_plot(ts_df.index, res.observed, res.seasonal, res.trend,
res.resid)
fig
```



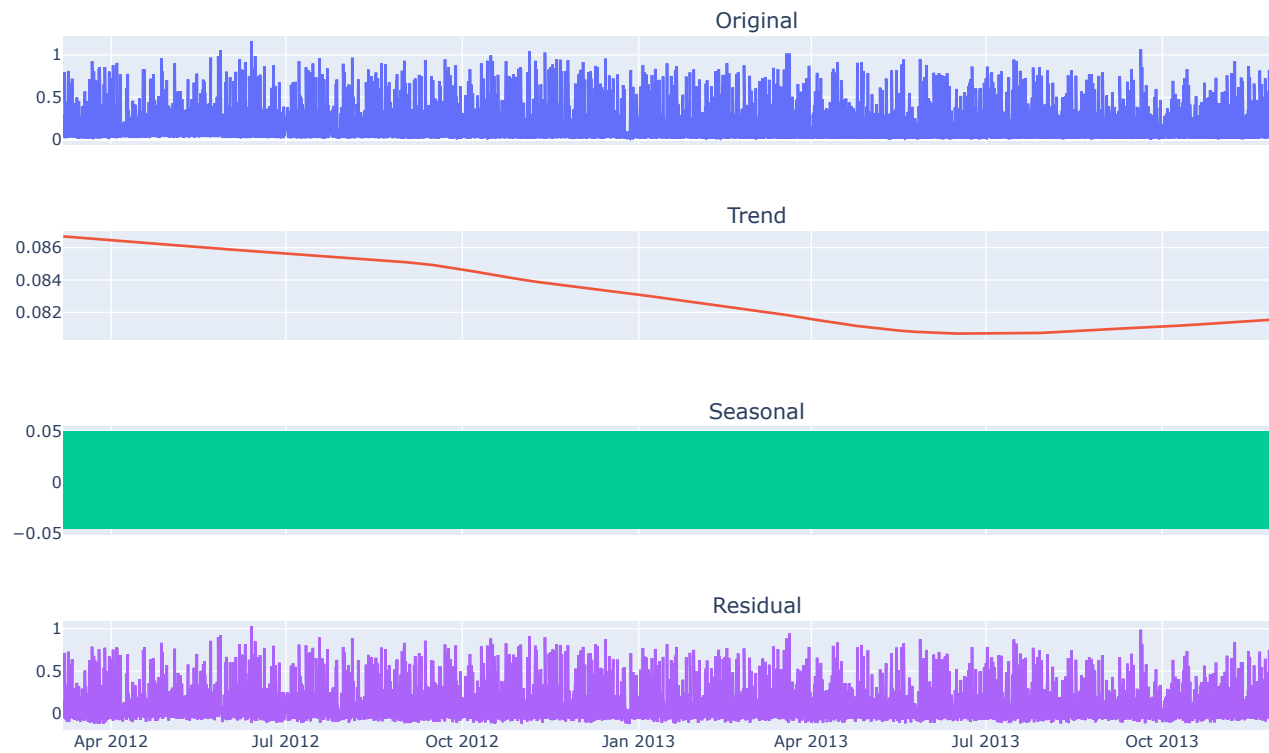
```
#Let's zoom in on a few days to better see the seasonality extracted
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])
fig
```



Seasonality and Trend Decomposition using Loess (STL)

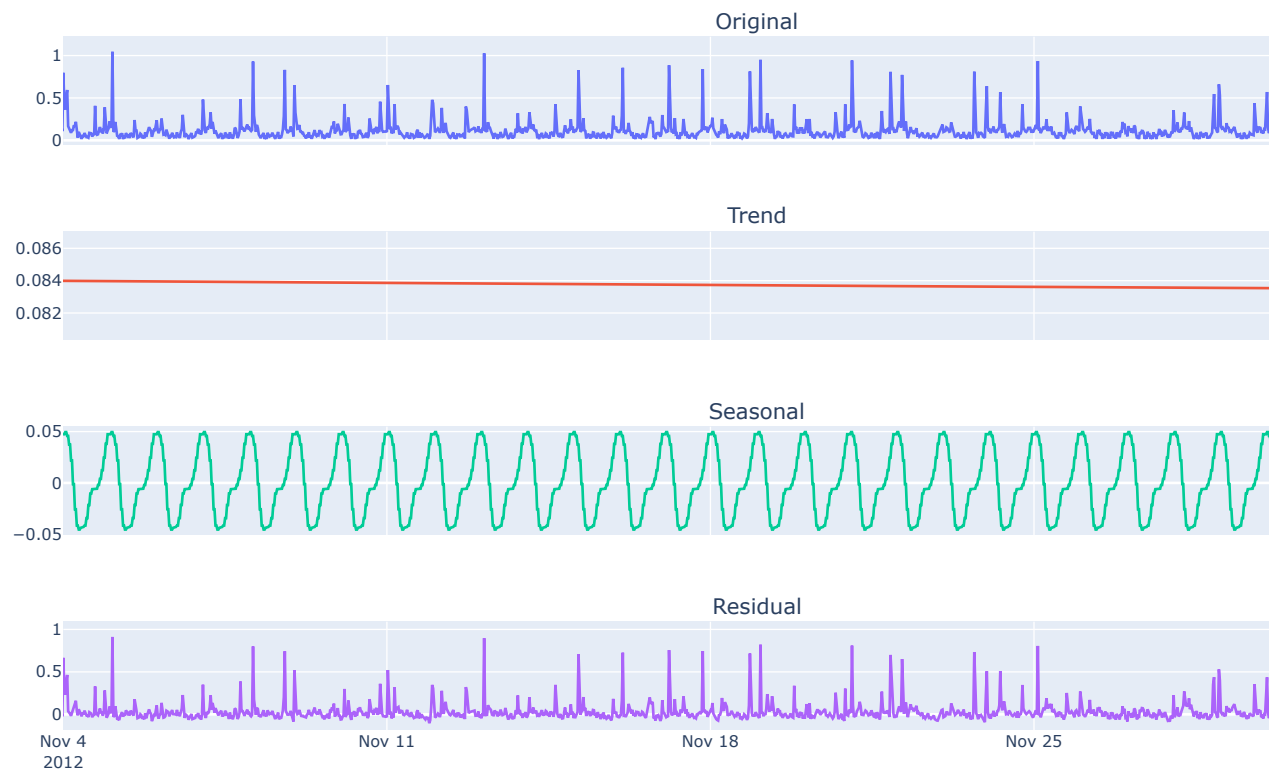
```
#Supports missing values and expects a series or dataframe with datetime index
stl = STL(seasonality_period=7*48, model = "additive")
res_new = stl.fit(ts_df.energy_consumption)
fig = res_new.plot(interactive=True, )
fig
```

Seasonal Decomposition



#Let's zoom in on a few days to better see the seasonality extracted
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])
fig

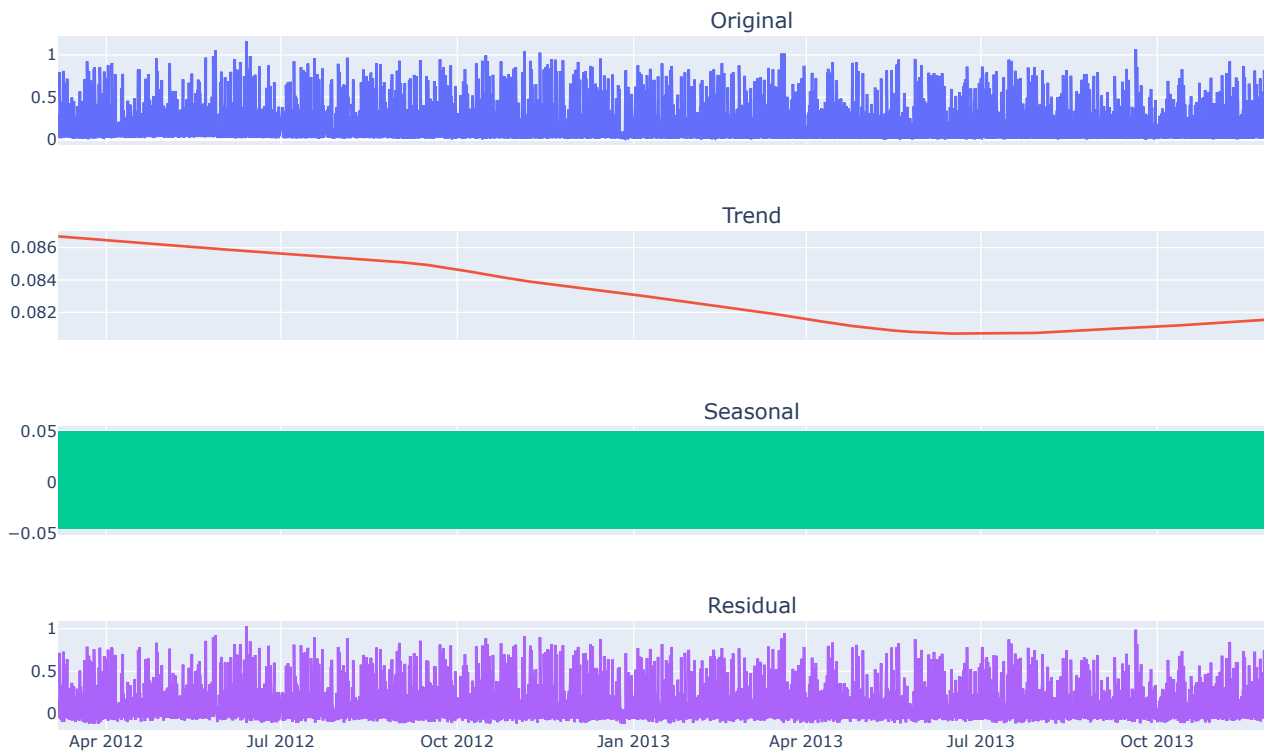
Seasonal Decomposition



Seasonality and Trend Decomposition using Loess and Fourier Terms (Fourier Decomposition)

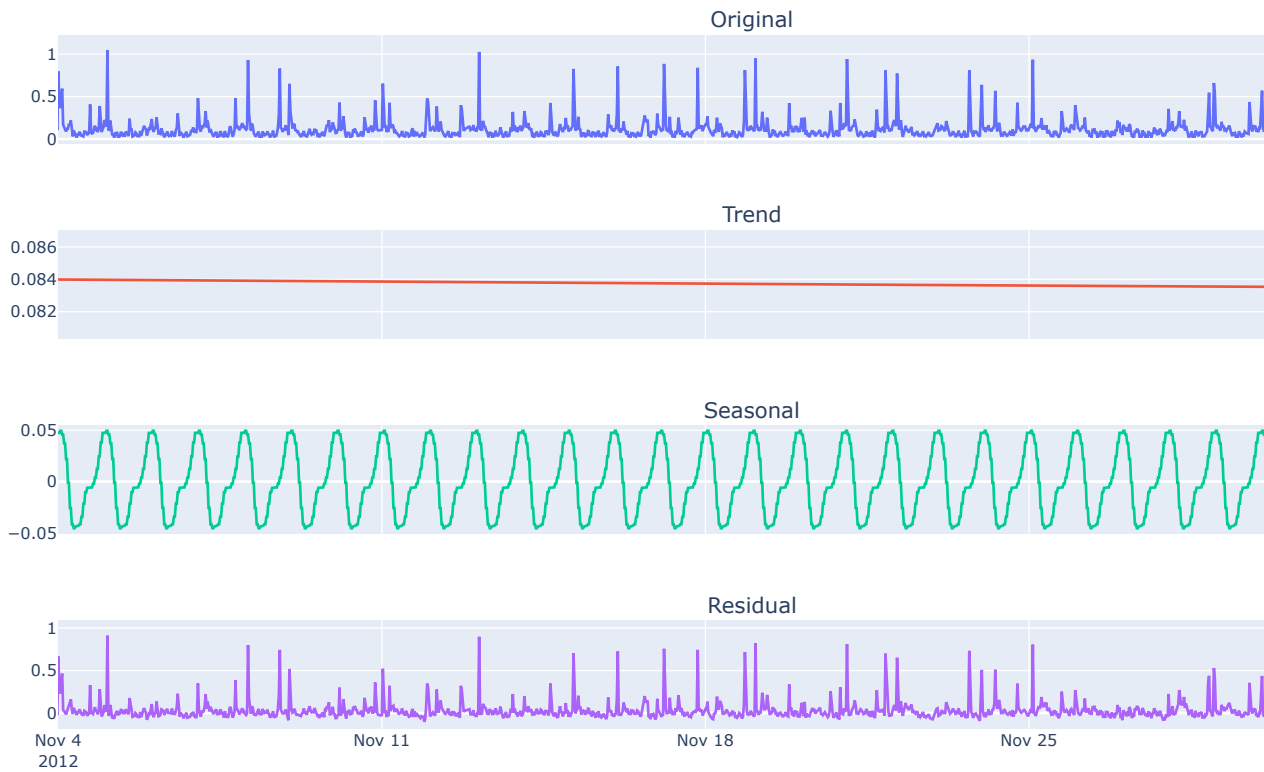
```
#Doesn't support missing values, and expects a series or datafeame with
datetime index
stl = FourierDecomposition(seasonality_period="hour", model = "additive",
n_fourier_terms=5)
res_new = stl.fit(pd.Series(ts.squeeze(), index=ts_df.index))
fig = res_new.plot(interactive=True)
fig
```

Seasonal Decomposition



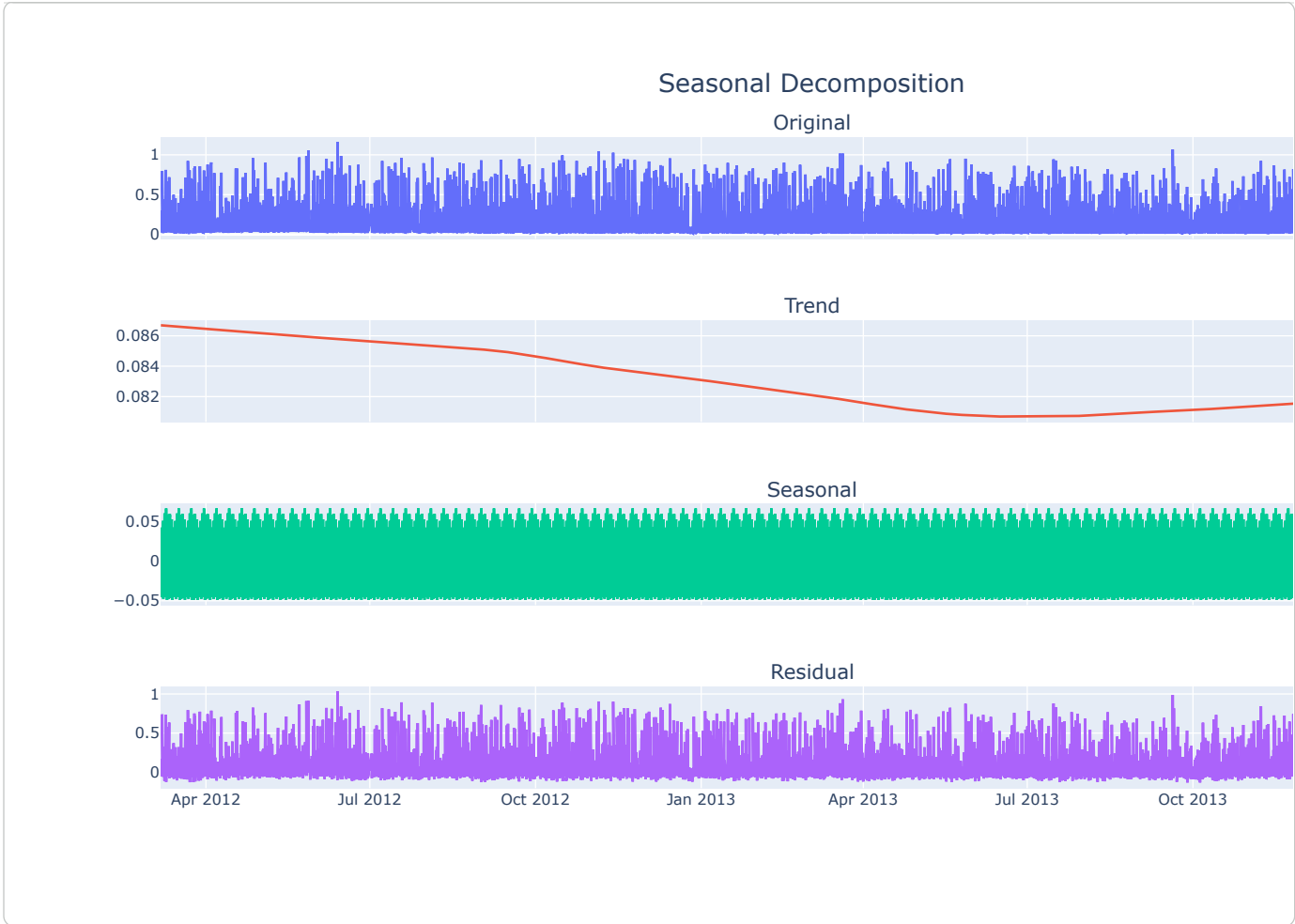
```
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])
fig
```

Seasonal Decomposition

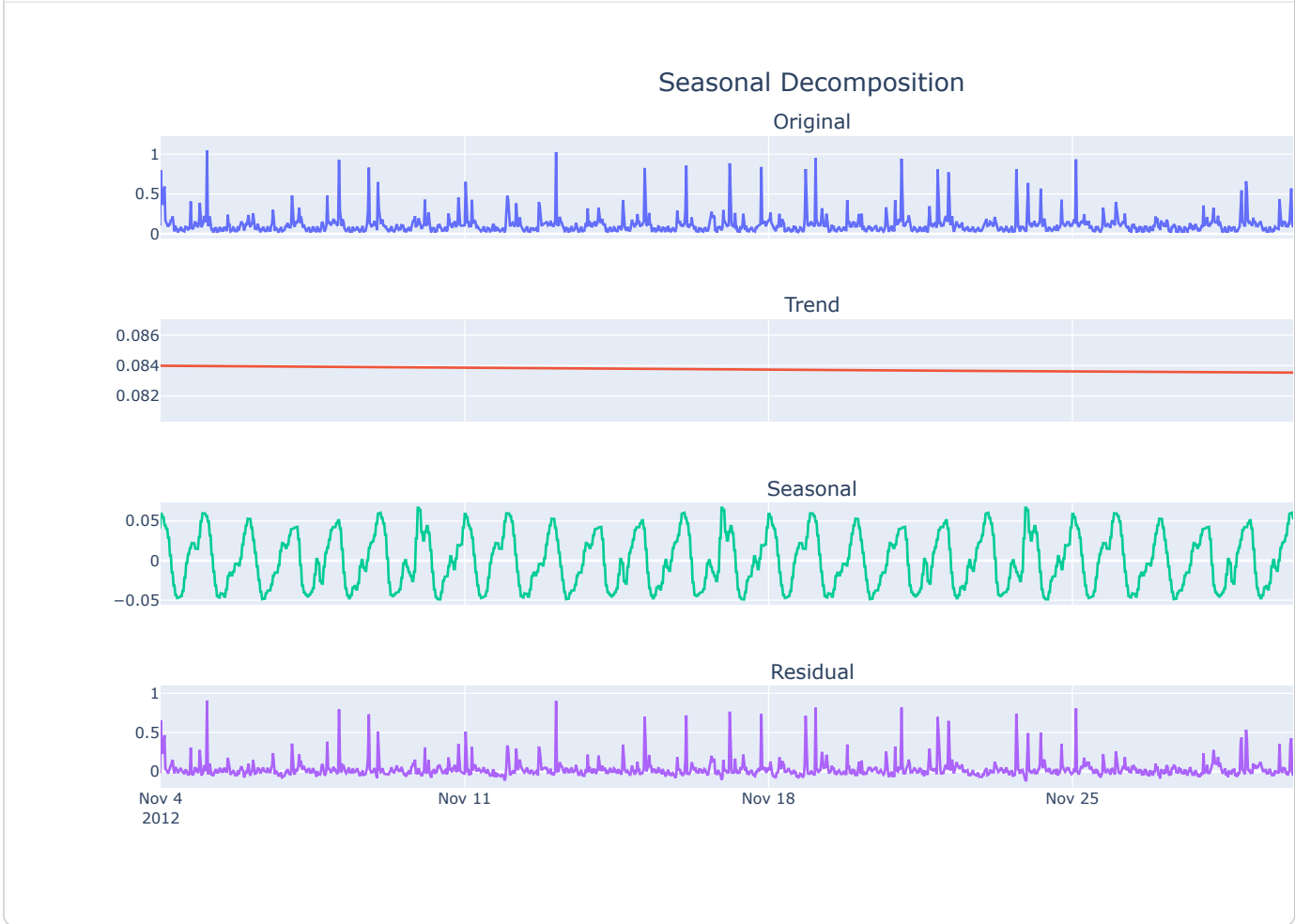


Custom Seasonality

```
stl = FourierDecomposition(model = "additive", n_fourier_terms=50)
res_new = stl.fit(pd.Series(ts, index=ts_df.index), seasonality=seasonality)
fig = res_new.plot(interactive=True)
fig
```



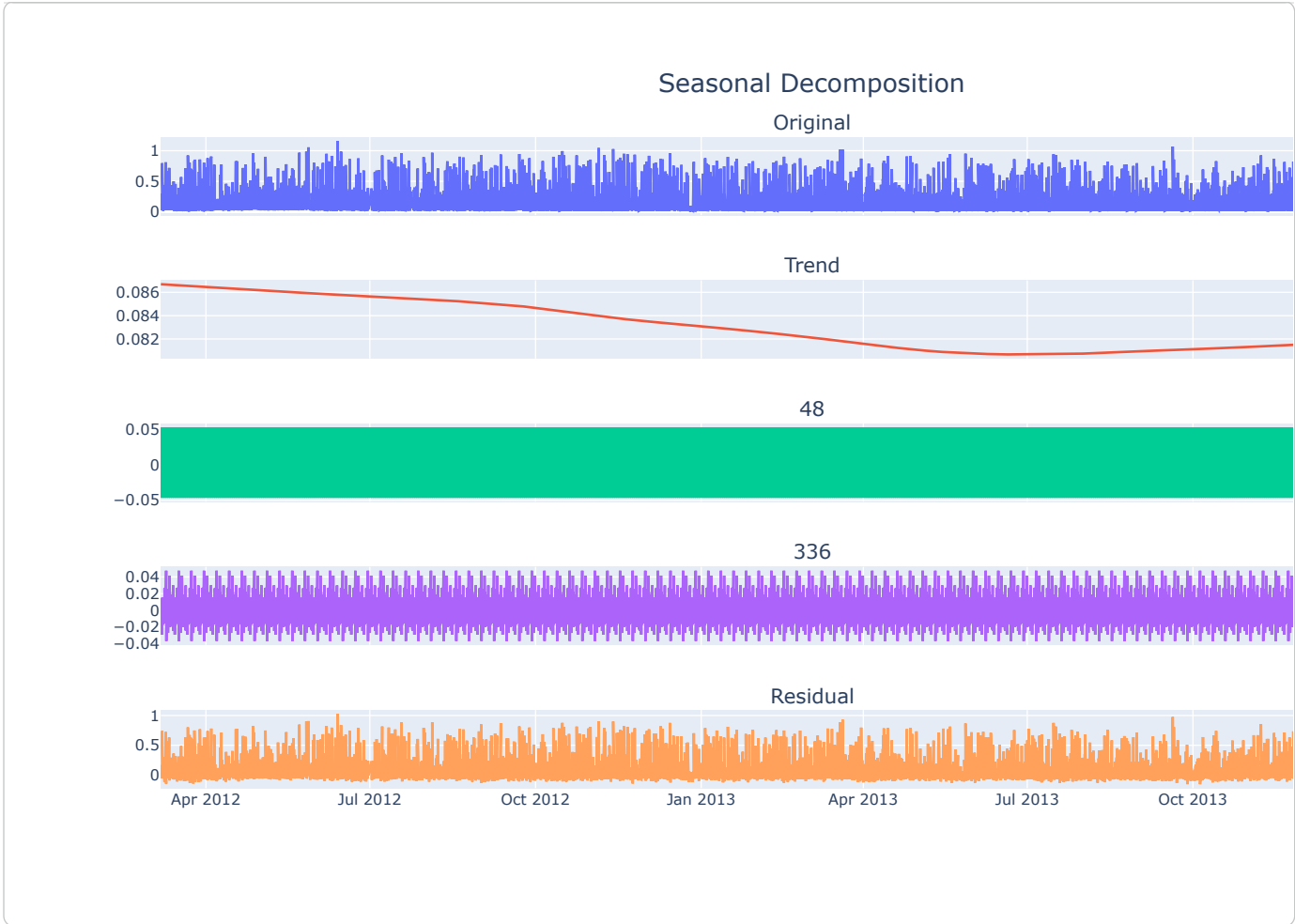
```
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])  
fig
```



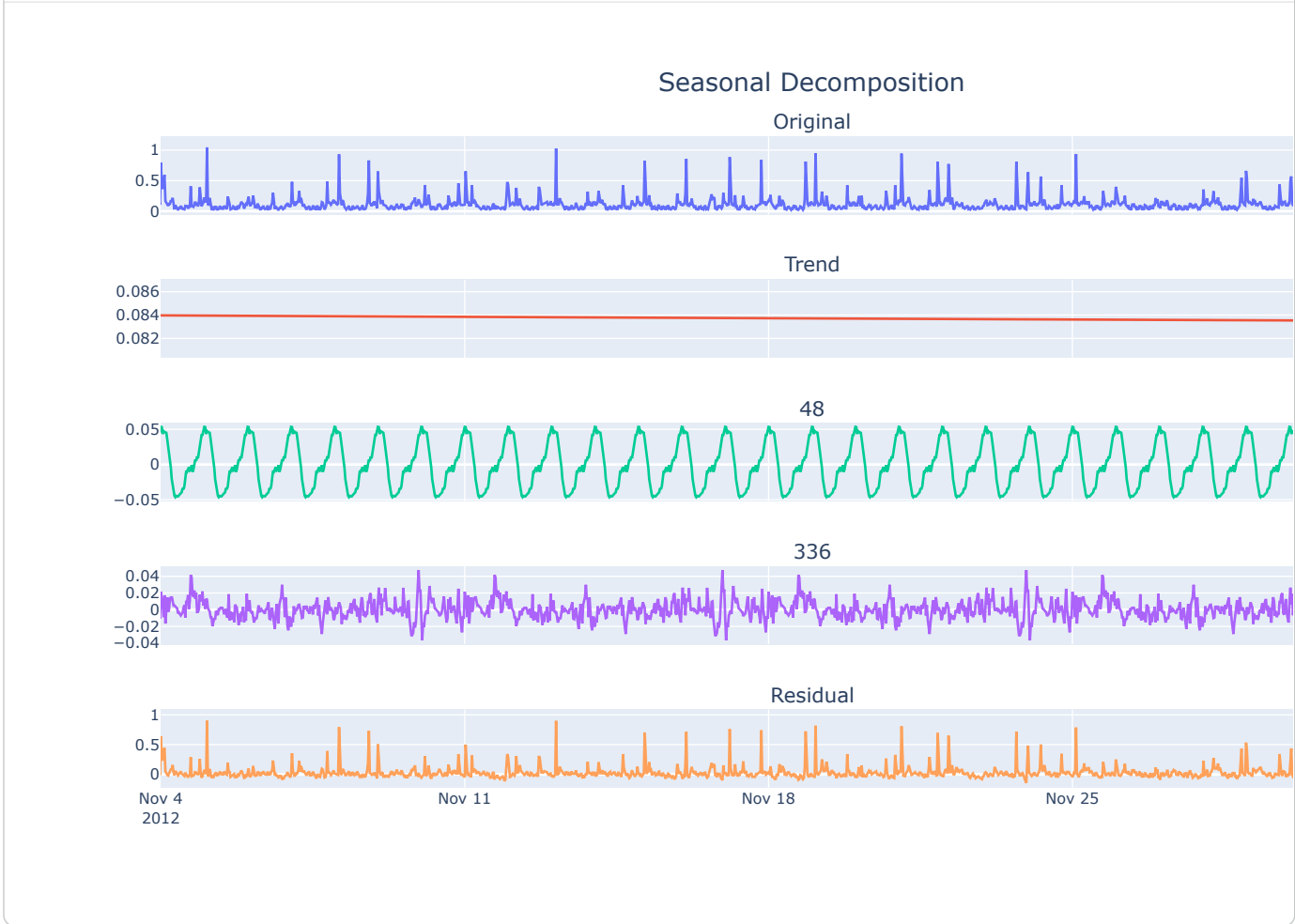
Multiple Seasonality Decomposition using Loess (MSTL)

Using Averages as the seasonal model

```
stl = MultiSeasonalDecomposition(seasonal_model="averages",seasonality_periods=[48*7, 48], model = "additive")  
res_new = stl.fit(pd.Series(ts, index=ts_df.index))  
fig = res_new.plot(interactive=True)  
fig
```

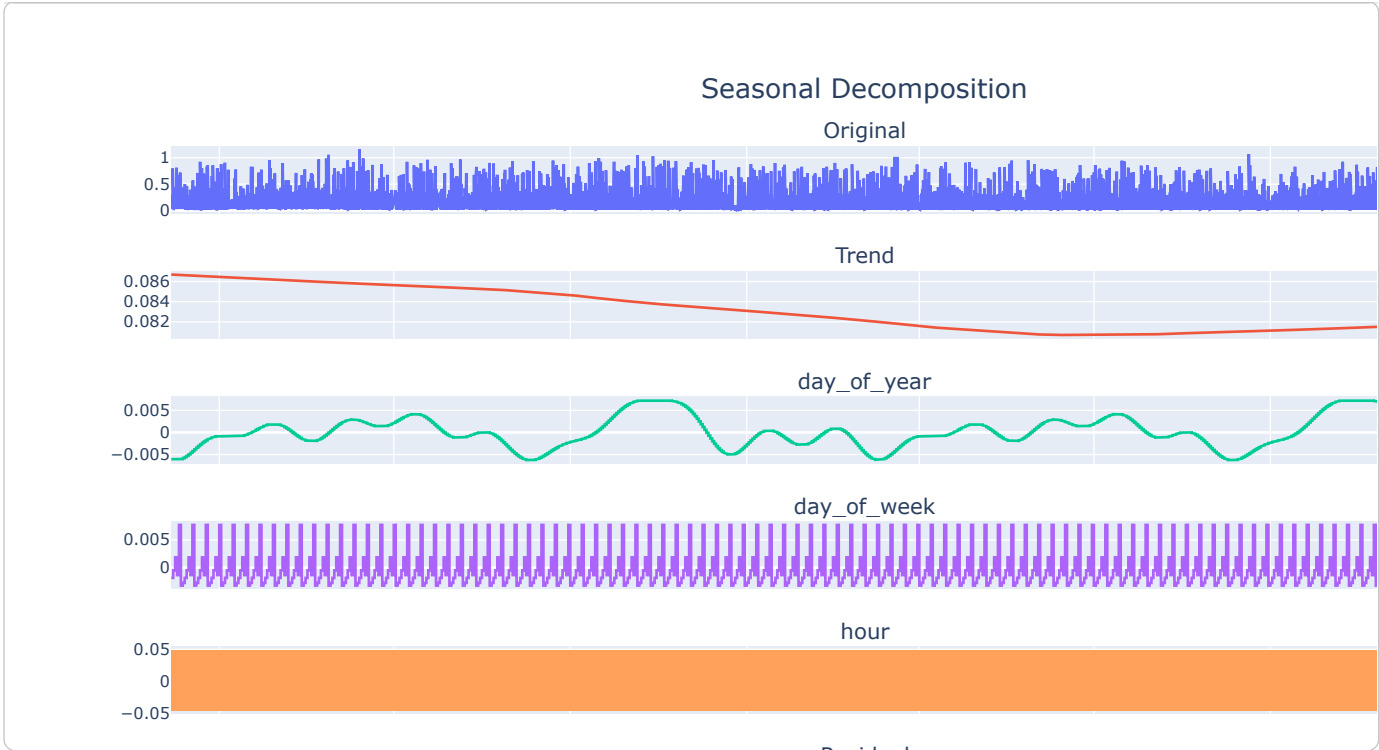


```
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])
fig
```



Using Fourier Decomposition as seasonal model

```
stl = MultiSeasonalDecomposition(seasonal_model="fourier",seasonality_periods=["day_of_year", "day_of_week", "hour"])
res_new = stl.fit(pd.Series(ts, index=ts_df.index))
fig = res_new.plot(interactive=True)
fig
```



```
fig.update_xaxes(type="date", range=["2012-11-4", "2012-12-4"])
fig
```

